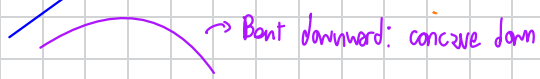
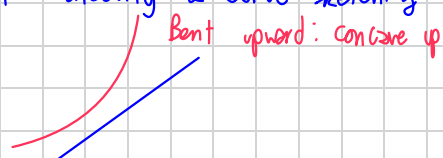
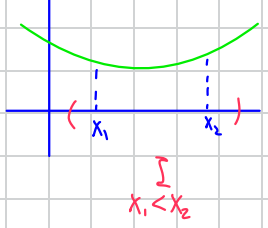


4.4 Concavity & Curve sketching



Concave up



$f(x)$ Increasing on I , then we say $f(x)$ is concave up in I

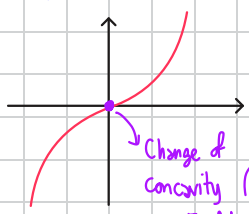
e.g. 1 $f(x) = x^2 \rightarrow f'(x) = 2x$ (increasing on $(-\infty, \infty)$)
 $\rightarrow f(x)$ is concave up on $(-\infty, \infty)$



Second Derivative test for concavity

Suppose $f''(x)$ exists on interval I , if $f''(x) > 0$ on I , then f' on $I \rightarrow f$ concave up
 if $f''(x) < 0$ on I , $f(x)$ concave down.

E.g. 3 $f(x) = x^3$



$$f'(x) = 3x^2, f''(x) = 6x \begin{cases} \oplus & x > 0 \\ \ominus & x < 0 \end{cases}$$

$$f(x) = \begin{cases} \cup & \text{on } (0, \infty) \\ \cap & \text{on } (-\infty, 0) \end{cases}$$

Change of concavity ($f''(x)=0$)
 $\rightarrow$ Inflection point

Def A point $(c, f(c))$ is an inflection point if the tangent line of $y = f(x)$ at $(c, f(c))$ exists, and the concavity changes at $(c, f(c))$

E.g. 4 $y = x^{2/3}$

$$f'(x) = \frac{1}{3}x^{-1/3}, f''(x) = -\frac{2}{9}x^{-4/3} = -\frac{2}{9x^{4/3}}$$

$f'(x)$ DNE when $x=0$

$$x^{2/3} = (x^{1/3})^2 = \text{Positive}$$

$$f'(0) = \frac{1}{0^+} = \infty$$

$(0,0) \rightarrow$ Inflection point

Q: why is an inflection pt. curve crosses T-line?

Fact 1: If f is diff / concave up at interval I then $\forall c \in I$, the graph of $y = f(x)$ stays above the tangent line passing through $(c, f(c))$
 i.e. $f(x) > f(c) + f'(c)(x-c); x \neq c, x \in I$

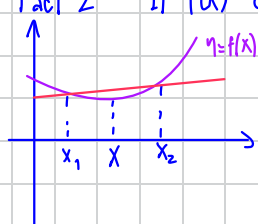
Proof Let $g(x) = f(x) - [f(c) + f'(c)(x-c)]$

$$\rightarrow g \text{ diff on } I, g(c) = 0, g'(x) = f'(x) - f'(c) \begin{cases} \ominus & (a, c) \\ \oplus & (c, b) \end{cases}$$



Fact 2 If $f(x)$ diff/concave up on I , then $\forall x_1, x_2 \in I$

$$f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1} (x - x_1) > f(x)$$



Let $g(x) = f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1} (x - x_1) - f(x)$

Observe: g diff on $[x_1, x_2]$

$$g(x_1) = 0 < 0$$

$$g'(x) = f'(c) - f'(x) \begin{cases} < 0 & x \in (x_1, x) \\ > 0 & c \in [x, x_2] \end{cases} \rightarrow g(x) > 0 \text{ on } (x_1, x_2)$$

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c)$$

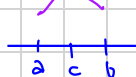
Necessary Condition for $(c, f(c))$ to be inflection pt:

Theorem: Suppose $(c, f(c))$ is an inflection pt. then either $f''(c) = 0$ or $f''(c)$ DNE

Proof: If $f''(c)$ DNE $\rightarrow$ done (Ex. $f(x) = x^{1/3}$ $f''(0)$ DNE and inflection pt)

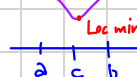
Suppose $f''(c)$ exists, since $(c, f(c))$ is an inflection pt, then $\exists$ neighborhood (a, b) of c such that

f' : $\text{loc max: } (f')' = 0$



Case 1

f' :



Case 2

Remark: The converse is untrue

(Counter example $f(x) = x^4 \rightarrow f''(x) = 12x^2$

$f''(0) = 0$ but no inflection pts)

Second Derivative Test for loc extrema

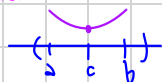
(When does C.P. give rise to loc extreme?)

Assume c is C.P. of f and f'' exist & continuous in nbhd (a, b) of $c \rightarrow f'(c) = 0$

Case 1

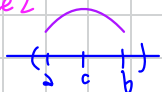
If $f''(c) > 0$, then $f''(x) > 0, x \approx c \rightarrow f$ concave up

$\therefore f(c)$ is the local min



Case 2

If $f''(c) < 0$, then $f(c)$ local max



Case 3 If $f''(c) = 0$

$f(x) = x^4 \rightarrow f''(0) = 0, x=0$ c.p. $\rightarrow f(0)$ loc min

$f(x) = -x^4 \rightarrow f''(0) = 0, x=0$ c.p. $\rightarrow f(0)$ loc max

$f(x) = x^3 \rightarrow f''(0) = 0, x=0$ c.p. $\rightarrow f(0)$ neither loc min nor loc max

} $\rightarrow$ INCONCLUSIVE

4.4/8 $y = 2 \cos X - \sqrt{2} X$; $-\pi \leq X \leq \frac{3\pi}{2}$

Step 1 $y' = 2 \sin X - \sqrt{2}$

$y'' = -2 \cos X$

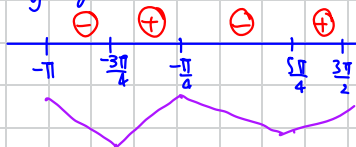
Step 2 Find all c.p. & plot number on X-axis

Set $y' = -2 \sin X - \sqrt{2} = 0$

$\sin X = -\frac{\sqrt{2}}{2}$

$X = -\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{5\pi}{4}$

Step 3 Study sign of f'



Loc min: $f(-\frac{3\pi}{4}), f(\frac{5\pi}{4})$, Loc max: $f(-\pi), f(-\frac{\pi}{4}), f(\frac{3\pi}{2})$

Step 4 Find all c s.t. $f''(c) = 0$

Set $f''(c) = -2 \cos X = 0 \rightarrow X = -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$

Step 5 : Study sign of f''



Concave up : $[-\pi, \frac{\pi}{2}] \cup [\frac{\pi}{2}, \frac{3\pi}{2}]$

Concave down : $[\frac{\pi}{2}, \frac{\pi}{2}]$

Inflection points : $(-\frac{\pi}{2}, f(-\frac{\pi}{2})), (\frac{\pi}{2}, f(\frac{\pi}{2}))$

4.4 (Continued)

Curve Sketching given $y = f(x)$

Recipe

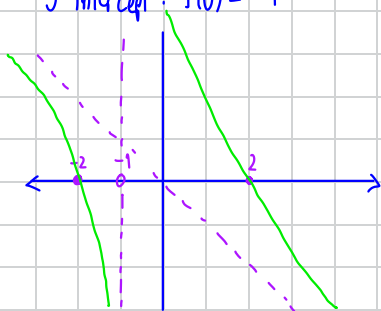
1. Understand D_f
2. f even/odd $\rightarrow$ symmetry about 0
symmetry about y -axis
3. x -intercept, y -intercept
4. Vertical/Horizontal/Oblique asymptote
5. Intervals on which f is increasing/decreasing
concave up/concave down

$$84 \quad y = \frac{4-x^2}{x+1}$$

Step 1 D_f : All x except $x = -1$

Step 2 $f(-x) = \frac{4-(-x)^2}{-x+1} = \frac{4-x^2}{1-x}$ (Neither even nor odd)
 $\neq f(x), \neq -f(x)$

Step 3 Solve $f(x) = 0$ to get x -intercepts $\rightarrow 4-x^2 = 0 \rightarrow x = \pm 2$
 y -intercept: $f(0) = 4$



Step 4 $\lim_{x \rightarrow -1^+} \frac{4-x^2}{x+1} = \frac{3}{0^+} = +\infty$ Heaven
 $\lim_{x \rightarrow -1^-} \frac{4-x^2}{x+1} = \frac{3}{0^-} = -\infty$ Hell

Horizontal $\lim_{x \rightarrow \infty} \frac{4-x^2}{x+1} = -\infty$
 $\lim_{x \rightarrow -\infty} \frac{4-x^2}{x+1} = \infty$

Oblique $y = x + 1$

Step 5: interval which $x \neq -1$

$$f''(x) = \frac{-(x+1)(2x+2) - (x^2+2x+4) \cdot 2(x+1)}{(x+1)^4}$$

$$= \frac{6}{(x+1)^3}$$

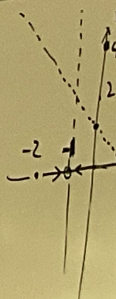
Oblique asymptote

Q. 1. $y = -x + 1$
Step 1: $f(x) = \frac{(x+1)(2x+2) - (x^2+2x+4)}{(x+1)^2}$
 $= \frac{-x^2-2x+4}{(x+1)^2}$
 $= \frac{-(x^2+2x+4)}{(x+1)^2}$

No c.p.? $\& f(x) = 0 \Rightarrow x^2+2x+4 = 0$ No roots.
Sign chart:
 $f: \begin{array}{ccc} & \ominus & \ominus \\ & \downarrow & \downarrow \\ & -1 & \end{array}$

$$f''(x) = -\frac{(x+1)(2x+2) - (x^2+2x+4) \cdot 2(x+1)}{(x+1)^4}$$

$$= -\frac{-3}{(x+1)^3} = \frac{3}{(x+1)^3}$$



4.5 Applied Optimization

E.g. A rectangle has perimeter fixed at L ($2x+2y=L$)

Q: when is the area A of rectangle maximized? $\rightarrow$ when it is a square ($x=y$)



$$y = \frac{829-2x}{2}$$

$$A = xy = \frac{x(829-2x)}{2} = \frac{829x}{2} - x^2 ; 0 \leq x \leq \frac{829}{2}$$

Want: Maximum of A on interval $[0, \frac{829}{2}]$

Step 1 C.P. in interior

$$0 = A'(x) = \frac{829}{2} - 2x \rightarrow x = \frac{829}{4} \text{ (in the interval)}$$

$$\text{Step 2 } A\left(\frac{829}{4}\right) = \frac{829}{2} \left(\frac{829}{4}\right) - \left(\frac{829}{4}\right)^2 = \left(\frac{829}{4}\right) \left(\frac{829}{2} - \frac{829}{4}\right) = \left(\frac{829}{4}\right)^2$$

$$A(0) = 0, A\left(\frac{829}{2}\right) = 0$$

Step 3 Choose largest A value $\rightarrow A_{\max} = \left(\frac{829}{4}\right)^2$

Economics Let $c(x)$ = cost of producing x units of a product

$r(x)$ = revenue

$p(x)$ = profit

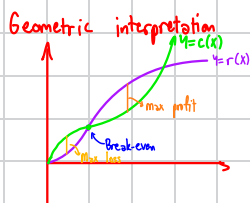
$$p(x) = r(x) - c(x); 0 \leq x \leq l$$

Want: Max $p(x)$ on $[0, l]$

If $p(x)$ is max at $x_0 \in (0, l)$, then $p'(x_0) = 0 \rightarrow r'(x_0) = c'(x_0)$

if x large

$c'(x)$: cost of producing 1 more unit after x units had been produced



Max loss/Max profit $\rightarrow r'(x_0) = c'(x_0)$

E.g. 6

-5 desks/day

-delivery cost of wood each time = \$5000

-storage cost = \$10/desk

Sol Let x be #days between deliveries $\rightarrow$ Amount of wood = $5x$

DAY 1 2 ... x-1 x

Units in Storage 5x-5 5x-10 ... 5 0

Storage Cost 50x-50 50x-100 ... 50 0

$$\rightarrow \text{Total storage cost per delivery cycle} = 50(x-1) + 50(x-2) + \dots + 50(1) + 0 = \frac{50(x)(x-1)}{2} = 25x^2 - 25x$$

$$\text{Total cost per delivery cycle} = 25x^2 + 5000$$

$$\text{Average daily cost} = \frac{25x^2 + 5000}{x} = 25x + \frac{5000}{x} ; 0 < x < \infty$$

Want: minimize $c(x) = 25x + \frac{5000}{x}$

C.P. $\rightarrow c'(x) \stackrel{!}{=} 0$

$$25 - \frac{5000}{x^2} = 0 \rightarrow x = \sqrt{200} = 10\sqrt{2}$$



$x = 10\sqrt{2}$ is global min of $c(x)$

4.6 Newton's method

Goal: Solve $f(x) = 0$

Idea $\rightarrow$ Numerical approximation

Wants x_1 from x_0

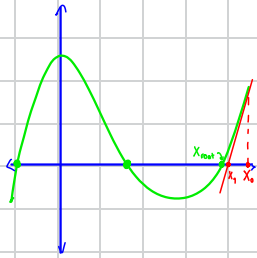
$$y = f(x_0) + f'(x_0)(x - x_0) \rightarrow \text{T-line at } x = x_0$$

$$f(x_0) + f'(x_0)(x - x_0) = 0$$

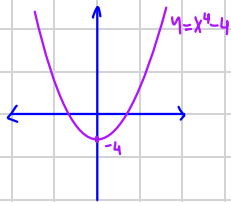
$$x - x_0 = -\frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$\text{In general for } n \geq 0 \rightarrow x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$



E.g. $\sqrt[4]{4} = ?$



Since $x^4 - 4 = 0$

$$x_0 = 1 \rightarrow x_1 = 1 - \frac{-3}{4(1)} = \frac{7}{4}$$

$$x_1 = \frac{7}{4} \rightarrow$$

$\vdots$

x_4 ใช้ x_5 ค่าที่ได้ก็ยังไม่ตรงนัก

$x^3 - 5x = 0$ ถ้า $x = -1, 1 \rightarrow$ กับ ถ้า x_n, x_{n+1} ไม่ตรงกัน

Cautions: 1) Avoid x_0 such that $f'(0) = 0$

2) Convergence to wrong root

3) Non-convergence of x_n 's

HW 4.4, 4.5, 4.6
+ Quiz